Counterexamples and Flux-Phase Structure for Signed Circulant Graphs

Anonymous manuscript

Abstract

For the signed circulant graph $C_n(1,2)$, a recent conjecture asserts that the alternating-triangle-flux class with negative Hamilton-cycle holonomy minimizes the spectral radius for every even $n \geq 8$. We disprove this conjecture and determine its smallest failure. Exact exhaustive computation verifies the conjectured bound for every even $8 \leq n \leq 30$, while an explicit signing at $n = 32$ has strictly smaller spectral radius; thus the finite minimality claim is computer-assisted. The counterexample is generated by a period-eight flux pattern that yields counterexamples for every $8 \mid n$ with $n \geq 32$. A Floquet reduction gives an exact quartic equation for the squared bands and the sharp period-eight value $\eta = 4 + \sqrt{10 + 2\sqrt{5}}$. We then explain the mechanism behind this phase. After squaring the signed operator, negative flux cancels all associated odd-displacement couplings, whereas positive flux creates localized defects. Closed-walk moments give a complete period-eight trichotomy: antipodal pairs of defects are exactly the sub-eight phases, the all-negative phase has squared radius eight, and every other phase lies above eight. For arbitrary period we derive the first three exact moments and obtain necessary defect-density and local-clustering inequalities. Finally, a complete exact classification shows that the period-eight phase is the unique minimizer, up to the natural operator equivalences, among periodic phases of primitive Hamilton-gauge period at most sixteen.

Contents

1 Introduction

Signed adjacency matrices provide a concrete meeting point of spectral graph theory, switching theory, and discrete magnetic or flux-phase ideas. If G is a graph and every edge receives a sign in $+1, -1$, switching the signs at a vertex conjugates the signed adjacency matrix by a diagonal sign matrix. Consequently the spectrum depends on cycle fluxes rather than on the chosen edge gauge. The optimization problem of minimizing spectral radius over all signings can therefore be viewed as a finite flux-phase problem.

We study this problem for the four-regular circulant

```
G_n=C_n(1,2),
```

whose vertices are the residues modulo n and whose edges join vertices at cyclic distances one and two. Suvagiya identified a distinguished family in which the triangle fluxes alternate [5]. For even n, the two members of that family with negative step-one Hamilton-cycle holonomy have spectral radius

```
rho_-(n)=2 sqrt(cos^2(pi/n)+cos^2(2pi/n)).
```

Conjecture 3 of [5] states that no signing of G_n has smaller spectral radius. That paper verified the assertion for $n = 8, 10, ..., 18$ and left the general lower bound open.

The inherited ingredients are the alternating-triangle family, its four switching/holonomy classes, the displayed formula for $\rho_-(n)$, and the conjecture itself [5]; the companion parity-family framework is developed in [4]. The counterexample, its minimality, the infinite periodic family, the sharp band edge, and the period-eight and low-period classifications proved here are new results of the present article.

The conjecture is false. Its failure is nevertheless delayed: every admissible order below 32 satisfies the proposed lower bound. The first counterexample is an order-32 signing whose triangle-flux word is the fourfold repetition of

```
(+,+,-,+,-,-,+,-).
```

The same word defines an eight-periodic operator. Its Floquet polynomial can be computed exactly, and a uniform positivity argument turns the isolated witness into an infinite family for all multiples of eight at least 32. The resulting analysis does more than refute the conjecture: it reveals a sharp period-eight spectral edge and a structural cancellation mechanism.

1.1 Main results

Our first result combines an exact witness with exhaustive finite exclusion.

Theorem A (smallest counterexample). The conjectured lower bound holds for every even n with $8 \leq n \leq 30$. It fails at $n = 32$. Hence 32 is the smallest counterexample order.

The exclusion through 30 is a finite computer-assisted theorem. Its proof enumerates a complete quotient of the switching classes and certifies every nonoptimizer by exact rational inequalities. The order-32 witness itself has a short exact positive-definiteness certificate.

The witness extends periodically.

Theorem B (infinite counterexample family). For every $n = 8L$ with $L \geq 4$ and either Hamilton holonomy, the period-eight signing defined in Section 4 satisfies

```
rho(A)^2 < 1561/200 < rho_-(n)^2.
```

Thus the conjecture fails at every multiple of eight at least 32. We do not claim that it fails at every even order above 32.

The Floquet analysis is sharp at infinite volume.

Theorem C (exact period-eight edge). For the target period-eight phase, the squared infinite-volume spectral radius is

```
eta=4+sqrt(10+2sqrt(5)),
```

and the top band reaches η only at Bloch parameter $z = 1$.

The local mechanism is described by a complete phase classification. For a legal eight-periodic flux word Q, let $D(Q)$ be the set of positive entries and let $R(Q)$ denote the squared infinite-volume spectral radius.

Theorem D (eight-barrier trichotomy). Exactly one of the following occurs:

```
D(Q)=emptyset:                R(Q)=8;
D(Q)={j,j+4}:               R(Q)=eta<8;
all other legal period-8 Q:    R(Q)>8.
```

In particular the target is the unique period-eight minimizer up to translation, reflection, and global edge-sign negation, and the all-negative phase is the unique runner-up.

The same squared-operator calculation gives information at every period. Put $d = |D(Q)|$, let a count adjacent positive pairs, and let b count positive pairs at cyclic distance two.

Theorem E (general-period moment obstruction). For every legal p-periodic phase,

```
M_1=4p,
M_2=20p+16d,
M_3=118p+168d+96a+48b.
```

If $R(Q) \leq 8$, then necessarily

```
d<=3p/4,
40d+96a+48b<=42p.
```

These are necessary conditions only.

Finally we classify a bounded but nontrivial periodic domain.

Theorem F (low-period frontier). Consider the infinite operators (2.1) whose Hamilton-gauge triangle word τ has least positive period at most 16, and minimize their squared infinite-volume radius R. The target phase is the unique minimizer up to translation, reflection, global edge-sign negation, and repetition of the unit cell. Repeated cells are retained only for certificate accounting and are identified spectrally by zone folding.

This theorem is also computer-assisted. The proof enumerates 2,626 legal flux/dihedral orbit representatives. A closed-walk hierarchy excludes 2,611 representations above the eight-barrier, one cancellation lemma treats eight repeated all-negative cells, five exact endpoint certificates handle the remaining competitors, and two displayed target rows are one phase related by zone folding.

1.2 Proof strategy

The first organizing device is Hamilton gauge. After switching all step-one edges to $+1$ except for a single holonomy cut, a signing is encoded by a periodic word τ of triangle fluxes. Its adjacent products

```
Q_i=tau_i tau_(i+1)
```

are the quadrilateral fluxes. Periodicity permits a Floquet decomposition into finite Hermitian Laurent matrices $H_Q(z)$. For the target word, the characteristic determinant is even in the eigenvalue and reduces to a quartic $P(y, c)$ in $y = \lambda^2$ and $c = z + z^{-1}$. An exact positive-coefficient expansion at the candidate edge proves the sharp bound and its equality case.

The second device is to square the infinite operator. In A_τ^2, every odd-displacement coefficient contains a factor $1 + Q_i$. Negative flux therefore cancels that coupling exactly, while positive flux activates an amplitude of absolute value two. Constant terms of even Bloch traces count signed closed walks. If all squared band values were at most eight, their successive moments would satisfy $M_{k+1} \leq 8M_k$. A positive excess therefore forces the spectrum above eight. This one-way implication, combined with defect geometry, proves the period-eight trichotomy and supplies the general period obstructions.

1.3 Scope and public status

The results distinguish three optimization domains. Theorem D concerns infinite-volume phases of displayed period eight. Theorem F concerns periodic phases of primitive Hamilton-gauge period at most 16. Theorem A concerns all finite signings only at the enumerated orders through 32. None of these results proves global optimality over all periods, all nonperiodic signings, or every finite order.

As of 20 August 2026, we found no direct public resolution of Conjecture 3 in the source paper, the author's listed follow-up materials, or the narrow direct-result searches recorded for this draft. This is a dated and bounded search statement, not an absolute priority claim.

1.4 Organization

Section 2 fixes switching, flux, Floquet, moment, and equivalence notation. Section 3 proves Theorem A. Sections 4 and 5 prove the periodic family and its sharp edge. Section 6 proves the eight-barrier trichotomy and explains the chiral symmetry. Section 7 derives the general-period moments and necessary conditions. Section 8 proves the bounded low-period frontier. Section 9 states the computer-assisted proof boundary and reproducibility model. Section 10 collects consequences and open problems. The appendices give the orbit-counting arguments, exact finite certificates, and computational protocol needed to audit the computer-assisted parts.

2 Signed Circulants and Flux Coordinates

2.1 Signed adjacency matrices and switching

For an integer $n \geq 8$, let $G_n = C_n(1, 2)$ have vertex set Z/nZ and edges

```
{i,i+1}, {i,i+2}    (i in Z/nZ).
```

A signing is a map $\sigma : E(G_n) -> +1, -1$. Its signed adjacency matrix A_σ is the real symmetric matrix whose ij entry is the sign of i, j when this is an edge and zero otherwise. We write

```
rho(A_sigma)=max{|lambda|: lambda in spec(A_sigma)}.
```

For a vertex-sign function $d : Z/nZ -> +1, -1$, let $D = \text{diag}(d_i)$. Replacing $\sigma_i j$ by $d_i \sigma_i j d_j$ replaces A_σ by $DA_\sigma D$ and therefore preserves the spectrum. This operation is switching.

Because G_n is connected and has $2n$ edges, its cycle space has dimension $2n - n + 1 = n + 1$. Hence there are exactly 2^{n+1} switching classes. We use a cycle-flux coordinate system adapted to the Hamilton cycle of step-one edges.

2.2 Hamilton gauge, triangle flux, and quadrilateral flux

Let a_i be the sign of the step-one edge $i, i + 1$ and b_i the sign of the step-two edge $i, i + 2$. Define the triangle flux and Hamilton holonomy by

```
tau_i=a_i a_(i+1) b_i,
alpha=product_(i=0)^(n-1) a_i.
```

The n triangle cycles together with the step-one Hamilton cycle form a cycle basis, so $(\tau_0, ..., \tau_{n-1}, \alpha)$ determines the switching class.

Switch so that all step-one edges are positive except possibly the edge crossing the cut from $n - 1$ to 0. Equivalently, extend a vector to the integer lattice with twisted boundary condition

```
x_(i+n)=alpha x_i.
```

The signed operator then has the local form

```
(A_tau x)_i=x_(i-1)+x_(i+1)+tau_(i-2)x_(i-2)+tau_i x_(i+2).        (2.1)
```

The distinguished local quadrilateral flux word is

```
Q_i=tau_i tau_(i+1).                                               (2.2)
```

For a p-periodic word τ, equation (2.2) implies $\prod_{i=0}^{p-1} Q_i = 1$. Conversely, any word $Q \in +1, -1^p$ with product $+1$ has exactly two periodic lifts, obtained by choosing $\tau_0 = +1$ or -1 and recursing via $\tau_{i+1} = Q_i \tau_i$.

We call

```
D(Q)={i:Q_i=+1}
```

the defect set and write $d = |D(Q)|$. We also use the cyclic local statistics

```
a=#{i:Q_i=Q_(i+1)=+1},
b=#{i:Q_i=Q_(i+2)=+1}.                                             (2.3)
```

When $n = 8$, the graph contains two additional step-two quadrilaterals not among the local cycles encoded by (2.2). This causes no loss of finite information: the cycle-basis coordinates (τ, α), rather than Q alone, parametrize every switching class and are the coordinates used in the all-signings enumeration.

2.3 Floquet matrices

Suppose τ has displayed period p. Write an integer index as $mp + r$, where $0 \leq r < p$, and seek Bloch vectors of the form

```
x_(mp+r)=z^m v_r.
```

Substitution in (2.1) gives a pxp Laurent matrix $H_\tau(z)$. More explicitly, for each transition from output residue r to an absolute source index j, write $j = mp + s$, $0 \leq s < p$, and add the transition coefficient times z^m to entry (r, s). This convention automatically adds coefficients when different transitions collide in short cells.

The identity

```
H_tau(z)^T=H_tau(z^(-1))                                          (2.4)
```

follows by reversing each undirected transition. Thus $H_\tau(z)$ is Hermitian on the unit circle.

For an infinite periodic phase, define

```
R(Q)=sup_(|z|=1) rho(H_tau(z))^2.                                  (2.5)
```

The two lifts of Q have the same value, so this notation is unambiguous. Notice that $R(Q)$ is always a **squared** spectral radius.

For every explicit fiber certificate below we use the canonical lift $\tau_0 = +1$, $\tau_{i+1} = Q_i\tau_i$, and the ordered fiber basis $(v_0, ..., v_{p-1})$. The other lift is obtained by the diagonal conjugacy and fiber reparametrization in Lemma 2.1, so it has the same squared spectral conclusions.

Whenever an explicit matrix representative is required, we write $H_Q(z) := H_{\tau^c an}(z)$ for the fiber formed from this canonical lift and basis. The notation $R(Q)$ and the moments $M_k(Q)$ remain lift-invariant.

If a finite graph has order $n = pL$ and Hamilton holonomy α, its cell sequence satisfies $u_{m+L} = \alpha u_m$. The unitary cell shift has eigenvalues exactly

```
z^L=alpha,                                                         (2.6)
```

and the finite matrix decomposes as

```
A_(pL,alpha)  ~=  direct_sum_(z^L=alpha) H_tau(z).                 (2.7)
```

Equation (2.6) is a discrete finite set. Equation $|z| = 1$ in (2.5) describes the infinite-volume spectrum. We will not substitute one for the other.

2.4 Operator equivalences

We record the equivalences used in classifying periodic phases.

Lemma 2.1 (translation, reflection, and edge-sign negation). Translating τ, reflecting it by $\tau_i- > \tau_{-i-2}$, or replacing τ by $-\tau$ preserves $R(Q)$. On flux words reflection is $Q_i- > Q_{-i-3}$.

Proof. Let $(T_r x)_i = x_{i-r}$ and $(Jx)_i = x_{-i}$. Direct substitution in (2.1) gives

```
T_r A_tau T_r^(-1)=A_(translated tau),
J A_tau J=A_(reflected tau).
```

Both are unitary conjugacies. For global edge-sign negation let $(Dx)_i = (-1)^i x_i$. The endpoints of a step-one edge have opposite D signs while those of a step-two edge have equal signs. Therefore

```
A_(-tau)=-D A_tau D.
```

The spectrum is negated and its square is unchanged. In an odd displayed cell the fiber coordinate changes from z to $-z$, but the full unit circle is preserved. ☐

Lemma 2.2 (zone folding). If τ has primitive period q and is displayed in a repeated cell $p = mq$, then

```
H_p(z) ~= direct_sum_(w^m=z) H_q(w).                                 (2.8)
```

In particular repetition of the unit cell preserves $R(Q)$.

Proof. Write a residue in the repeated cell as $r + kq$, where $0 \le r < q$ and $0 \le k < m$. Decompose the mq-dimensional fiber according to the unitary internal translation by q. On its w-eigenspace vectors have the form

```
x_(r+kq)=w^k v_r.
```

The repeated boundary condition is equivalent to $w^m = z$. Since all coefficients in (2.1) are q-periodic, every transition factors out w^k and the remaining action on v is exactly $H_q(w)$. The m roots of $w^m = z$ give mutually orthogonal eigenspaces and their dimensions sum to mq. This proves (2.8), including multiplicities. □

2.5 Closed-walk moments

For a periodic phase define

```
M_k(Q)=CT_z tr(H_Q(z)^(2k)),
F_k(Q)=M_(k+1)(Q)-8M_k(Q).                                           (2.9)
```

Constant-term extraction equals normalized integration around the unit circle:

```
M_k(Q)=(1/(2pi)) integral_0^(2pi)
       sum_j lambda_j(e^(i theta))^(2k) dtheta.                       (2.10)
```

It is also the signed sum of closed walks of length $2k$ per period cell.

Lemma 2.3 (moment barrier). If $R(Q) \le 8$, then $M_{k+1}(Q) \le 8M_k(Q)$ for every $k \ge 1$. Equivalently,

```
F_k(Q)>0  ==>  R(Q)>8.                                               (2.11)
```

Proof. Under $R(Q) \le 8$, every $y_j(\theta) = \lambda_j(e^{i\theta})^2$ lies in $[0, 8]$. Hence $y_j^{k+1} \le 8y_j^k$. Sum over j and integrate using (2.10). □

We emphasize that (2.11) is one-way. A nonpositive excess, or any finite list of nonpositive excesses, does not prove $R(Q) \le 8$.

2.6 Distinguished constants

For even $n \ge 8$, the conjectured threshold inherited from [5] is

```
rho_-(n)=2 sqrt(cos^2(pi/n)+cos^2(2pi/n)),
rho_-(n)^2=4+2cos(2pi/n)+2cos(4pi/n).                                 (2.12)
```

The target period-eight squared spectral edge is

```
eta=4+sqrt(10+2sqrt(5)),
rho_*=sqrt(eta).                                                       (2.13)
```

These quantities play different roles: $\rho_-(n)$ is the finite conjectured lower bound, while η is the exact infinite-volume squared radius of the counterexample phase.

3 The Smallest Counterexample

We now prove Theorem A. The proof has two logically separate parts. First, an explicit order-32 signing is certified by exact matrix inequalities. Second, a complete finite enumeration excludes every admissible order below 32. The first part is short enough to give here; the quotient and certificate coverage for the second part is proved in Appendix A.

3.1 The order-32 signing

Let $n = 32$, choose Hamilton holonomy $\alpha = +1$, and repeat the triangle-flux word

```
tau=(+,+,-,+,-,-,+,-)                                            (3.1)
```

four times. In Hamilton gauge all step-one signs are positive and the step-two sign at $i, i+2$ is τ_i. The resulting quadrilateral flux is

```
Q=(+,-,-,-)^8.                                                    (3.2)
```

Let A_32 denote the resulting signed adjacency matrix.

Proposition 3.1. The matrix A_32 satisfies

```
rho(A_32)^2 < 1561/200 < rho_-(32)^2.                             (3.3)
```

Proof. The eight-site Floquet calculation in Section 4 gives the explicit polynomial $P(y, c)$ in (4.5). Put $B = 1561/200$, $u = y - B$, and $t = 2 - c$. Equation (4.6) displays $P(B + u, 2 - t)$ as a polynomial whose coefficients are nonnegative and whose constant term is $84332641/1600000000 > 0$. Hence $P(y, c) > 0$ for every $y \geq B$ and $c \in [-2, 2]$. Every squared eigenvalue of a unit-circle fiber is a nonnegative root of P; therefore every fiber has squared radius strictly less than B. The finite decomposition (4.3), with $L = 4$ and $\alpha = +1$, gives $\rho(A_32)^2 < 1561/200$.

For the other inequality, equation (2.12) gives

```
rho_-(32)^2
 =4+sqrt(2+sqrt(2))+sqrt(2+sqrt(2+sqrt(2))).                      (3.4)
```

This algebraic number is the unique real root in $(7809/1000, 781/100)$ of

```
X^8-32X^7+432X^6-3216X^5+14456X^4
 -40224X^3+67736X^2-63184X+25022.                                 (3.5)
```

A Sturm sequence has variation difference one at the two endpoints, and exact substitution of the nested radical identifies the isolated root. Since

```
1561/200 < 7809/1000,
```

the second inequality in (3.3) follows. □

The witness is not gauge-dependent. For example, choosing step-one signs

```
(+,-,+,-,-,+,-,+)^4
```

and solving $b_i = \tau_i a_i a_{i+1}$ gives step-two signs

```
(-,-,+,+,+,+,-,-)^4.
```

The diagonal switching vector $(+, +, -, -, +, -, -, +)^4$ conjugates this matrix to A_32. This supplies an independent reconstruction from flux data rather than from the stored edge list.

3.2 Exhaustive exclusion below 32

The conjecture is posed exactly for even $n \geq 8$; hence the admissible orders below 32 are

```
8,10,12,14,16,18,20,22,24,26,28,30.                               (3.6)
```

Proposition 3.2 (finite exclusion). For every order in (3.6) and every signing σ of $C_n(1, 2)$,

```
rho(A_sigma)>=rho_-(n).                                           (3.7)
```

This is a finite computer-assisted proposition.

Proof. Switching classes are represented in the cycle coordinates (τ, α). Global edge-sign negation and the dihedral action are used only after their spectral invariance has been proved. For $n \leq 20$, all 2^{n+1} classes are enumerated directly. For $n = 22$, and for the production orders $24, 26, 28, 30$, the legal quadrilateral words are partitioned into dihedral bracelets, with both holonomies retained. Appendix A proves that the orbit sizes sum to 2^{n+1} and gives an independent Burnside count.

For every representative other than the distinguished optimizer, the enumeration stores a nonzero rational vector v and verifies exactly

```
(v^T A_sigma^2 v)/(v^T v) >= rho_-(n)^2.                         (3.8)
```

The trigonometric threshold in (3.8) is compared as a real algebraic number; floating eigenvalues are used only to propose v. The distinguished class is checked by its exact known spectrum. Thus every switching class satisfies (3.7). The state counts and certificate totals in Table 3.1 agree with the complete quotient counts, and there are no unresolved representatives. □

n	represented switching classes	exact nonoptimizer certificates
8	512	510
10	2,048	2,046
12	8,192	8,190
14	32,768	32,766
16	131,072	131,070
18	524,288	524,286
20	2,097,152	2,097,150
22	8,388,608	97,467 quotient certificates
24	33,554,432	353,811 quotient certificates
26	134,217,728	1,299,063 quotient certificates
28	536,870,912	4,810,471 quotient certificates
30	2,147,483,648	17,929,599 quotient certificates

The last column counts spectral representatives, not switching classes. The quotient multiplicities in Appendix A account exactly for the difference.

3.3 Proof of Theorem A

Proposition 3.2 excludes every admissible order below 32. Proposition 3.1 gives a strict counterexample at 32. Since the domain consists of even integers at least eight, no admissible order lies between 30 and 32. Therefore 32 is the smallest counterexample order. □

The theorem does not assert that the order-32 witness is unique, nor does it assert failure at every even order above 32.

4 Periodic Construction and Floquet Reduction

We now place the order-32 pattern in an arbitrary number of eight-site cells. Let

```
tau_*=(+,+,-,+,-,-,+,-)                                        (4.1)
```

and impose either Hamilton holonomy $\alpha \in +1, -1$ on a graph of order $n = 8L$.

4.1 The eight-site fiber

Write $i = 8m + r$, $0 \le r < 8$, and put $x_{8m+r} = z^m v_r$. Substitution in (2.1) gives

```
H(z)=
[ 0  1  1  0  0  0   z^-1   z^-1 ]
[ 1  0  1  1  0  0    0    -z^-1 ]
[ 1  1  0  1 -1  0    0      0   ]
[ 0  1  1  0  1  1    0      0   ]
[ 0  0 -1  1  0  1   -1      0   ]
[ 0  0  0  1  1  0    1     -1   ]
[ z  0  0  0 -1  1    0      1   ]
[ z -z  0  0  0 -1    1      0   ].                        (4.2)
```

On $|z| = 1$, replacing z by its complex conjugate z^{-1} transposes the matrix, so (4.2) is Hermitian.

Proposition 4.1 (finite Floquet decomposition). For $n = 8L$,

```
A_(8L,alpha) ~= direct_sum_(z^L=alpha) H(z).               (4.3)
```

Proof. Let S be the unitary shift on the L cell coordinates with twisted boundary $u_{m+L} = \alpha u_m$. Its normalized eigenvectors are $L^{-1/2}(1, z, ..., z^{L-1})$ with $z^L = \alpha$. Distinct roots are orthogonal by the geometric-sum identity. The signed operator is a finite sum of residue matrices tensored with powers of S, so each eigenspace is invariant and its restriction is (4.2). There are L such eight-dimensional eigenspaces, and their dimensions sum to $8L$. □

Thus the finite problem uses the discrete set $z^L = \alpha$, while the infinite periodic problem uses every $|z| = 1$.

4.2 Exact determinant

Let x be the spectral parameter. Fraction-free elimination of $xI - H(z)$ gives

```
det(xI-H(z))
 =x^8-16x^6+(80-2(z+z^-1))x^4
   +(-128+16(z+z^-1))x^2
   +(z^2+z^-2)-13(z+z^-1)+40.                             (4.4)
```

Set

```
y=x^2,        c=z+z^-1.
```

Since $z^2 + z^{-2} = c^2 - 2$, equation (4.4) becomes

```
P(y,c)=y^4-16y^3+(80-2c)y^2+(-128+16c)y+c^2-13c+38.       (4.5)
```

For $|z| = 1$, $c = 2\cos(\theta)$ lies in $[-2, 2]$. Hermiticity implies that every eigenvalue λ is real, and (4.5) shows that λ^2 is a nonnegative root of $P(y, c)$.

4.3 A uniform rational bound

Put

```
B=1561/200,      u=y-B,       t=2-c.
```

Direct substitution in (4.5) gives

```
P(B+u,2-t)
 =u^4+(761/50)u^3+2u^2t+(1337363/20000)u^2
  +(761/50)ut+(136311081/2000000)u
  +t^2+(119121/20000)t+84332641/1600000000.               (4.6)
```

Every coefficient in (4.6) is nonnegative and the constant term is positive. Therefore $P(y,c) > 0$ whenever $y \geq B$ and $c \leq 2$. In particular, no squared eigenvalue of any unit-circle fiber reaches B; hence

```
rho(H(z))^2<B                                                    (4.7)
```

for every $|z| = 1$. Proposition 4.1 now yields

```
rho(A_(8L,alpha))^2<1561/200                                   (4.8)
```

for both holonomies and every $L \geq 1$.

4.4 Comparison with the conjectured threshold

At $n = 32$, the Sturm isolation (3.4)-(3.5) gives

```
rho_-(32)^2>7809/1000>1561/200.                              (4.9)
```

For $n \geq 32$, both angles $2\pi/n$ and $4\pi/n$ lie in $(0, \pi)$ and decrease as n increases. Since cosine decreases with its argument on $(0, \pi)$, equation (2.12) shows that $\rho_-(n)^2$ increases with n. Thus

```
rho_-(n)^2>=rho_-(32)^2>1561/200.                            (4.10)
```

Combining (4.8) and (4.10) proves Theorem B for $n = 8L$, $L \geq 4$. ☐

The restriction to multiples of eight comes from the construction, not from the threshold comparison. No conclusion is drawn here for other even orders.

5 The Exact Period-Eight Spectral Edge

The rational number in Section 4 is convenient for proving strict finite counterexamples, but it is not the true band edge. We now prove Theorem C.

5.1 The endpoint roots

At $c = 2$, equation (4.5) becomes

```
P(y,2)=y^4-16y^3+76y^2-96y+16.                              (5.1)
```

Translate $y = x + 4$. Then

```
P(x+4,2)=x^4-20x^2+80.                                        (5.2)
```

Putting $w = x^2$ gives $w^2 - 20w + 80 = 0$, whose roots are $10 \pm 2\sqrt{5}$. The four real roots of (5.1), in increasing order, are

```
4-sqrt(10+2sqrt(5)),
4-sqrt(10-2sqrt(5)),
4+sqrt(10-2sqrt(5)),
4+sqrt(10+2sqrt(5)).                                           (5.3)
```

Thus the largest endpoint root is

```
eta=4+sqrt(10+2sqrt(5)).                                      (5.4)
```

5.2 Sharp positivity and uniqueness

Set

```
s=sqrt(10+2sqrt(5)),       eta=4+s,
u=y-eta,                   t=2-c.
```

Substitution in (4.5) yields the identity

```
P(eta+u,2-t)
 =u^4+4s u^3+2u^2t+(40+12sqrt(5))u^2
  +4s ut+8sqrt(5)s u+t^2+(4sqrt(5)-3)t.                  (5.5)
```

All coefficients are positive. Hence for $u, t \geq 0$, the right side is nonnegative, and it vanishes only at $u = t = 0$. Since every unit-circle fiber has $c \leq 2$, equation (5.5) excludes every squared eigenvalue above η and permits equality only when $c = 2$.

At $c = 2$, equation (5.1) has η as a root, so $\pm\sqrt{\eta}$ are eigenvalues of $H(1)$. Therefore

```
sup_(|z|=1) rho(H(z))^2=eta.                             (5.6)
```

On the unit circle, $z + z^{-1} = 2$ if and only if $z = 1$. This proves both the value and the uniqueness assertion in Theorem C. □

For later use we also record that η has minimal polynomial

```
Y^4-16Y^3+76Y^2-96Y+16                                  (5.7)
```

Indeed, translation $Y = X + 4$ reduces (5.7) to $X^4 - 20X^2 + 80$. By Gauss's lemma, a rational factorization may be taken into monic integral quadratics. Vanishing of the cubic and linear coefficients reduces it either to $(X^2 + a)(X^2 + b)$, where $a + b = -20$ and $ab = 80$, or to $(X^2 + rX + s)(X^2 - rX + s)$, where $s^2 = 80$. The first case would make a, b roots of $T^2 + 20T + 80$, whose discriminant 80 is not a square; the second has no rational s. Thus (5.7) is irreducible over Q. The number η lies in $(1951/250, 1561/200)$; in particular $\eta < 8$.

5.3 Monotonicity of the top band

Let $r(c)$ be the largest real root of $P(y, c)$. At $c = -2$,

```
P(y,-2)=(y^2-12y+34)(y^2-4y+2),
r(-2)=6+sqrt(2)=:y_0.                                    (5.8)
```

For $-2 < c \leq 2$, substitution gives

```
P(y_0,c)=(c+2)(c+5-8sqrt(2))<0.                         (5.9)
```

Since $P(y, c)$ tends to positive infinity with y, (5.9) implies $r(c) > y_0$ for $c > -2$.

Moreover,

```
P_c(y,c)=2(c-c_0(y)),        c_0(y)=y^2-8y+13/2.         (5.10)
```

The function c_0 is increasing for $y \geq y_0 > 4$, and

```
c_0(y_0)=-7/2+4sqrt(2)>2.
```

Thus $P_c(y, c) < 0$ throughout $y \geq y_0$ and $c \leq 2$. If $-2 \leq c_1 < c_2 \leq 2$ and $y_1 = r(c_1)$, then $P(y_1, c_2) < P(y_1, c_1) = 0$; hence $P(., c_2)$ has a root above y_1. Therefore $r(c)$ is strictly increasing on $[-2, 2]$.

5.4 Finite holonomies

For $\alpha = +1$, the finite set $z^L = 1$ contains $z = 1$. By (5.6),

```
rho(A_(8L,+1))^2=eta.                                    (5.11)
```

For $\alpha = -1$, the admissible parameters are $z_k = exp((2k+1)\pi i/L)$. Their largest c value is $2\cos(\pi/L)$, which is strictly less than two. The monotonicity just proved gives

```
rho(A_(8L,-1))^2=r(2cos(pi/L))<eta.                      (5.12)
```

As L tends to infinity, $2\cos(\pi/L)$ tends to two, and continuity of the eigenvalues of the Hermitian fiber gives convergence of (5.12) to η. Thus both holonomy sectors have the same infinite-volume edge, but only the positive sector attains it at each finite size.

6 The Eight-Barrier and Structural Optimum

We now prove Theorem D without relying on the list of 17 individual competitor certificates. The proof begins with an exact local identity for the squared operator and ends with a finite four-case analysis of two-defect geometry.

6.1 Squaring the operator

Starting from (2.1), multiply the four transitions twice. The complete row of A_τ^2 from residue i is

displacement	coefficient
-4	$Q_{i-4}Q_{i-3}$
-3	$\tau_{i-3}(1+Q_{i-3})$
-2	1
-1	$\tau_{i-2}(1+Q_{i-2})$
0	4
$+1$	$\tau_{i-1}(1+Q_{i-1})$
$+2$	1
$+3$	$\tau_i(1+Q_i)$
$+4$	Q_iQ_{i+1}

For completeness, consider for example displacement $+3$. It arises from the two paths $+1, +2$ and $+2, +1$, with weights τ_{i+1} and τ_i. Since $\tau_{i+1} = \tau_i Q_i$, their sum is $\tau_i(1+Q_i)$. The other odd displacements follow by translation and reversal. Displacement $+4$ has the single path $+2, +2$, of weight $\tau_i\tau_{i+2} = Q_iQ_{i+1}$. The diagonal has four immediate reversals, the ± 2 displacements have their surviving unit coefficient after cancellation of the remaining routes, and reversal gives the negative displacements.

Thus a negative flux $Q_i = -1$ cancels the associated odd-displacement couplings, while a positive flux opens them with absolute amplitude two. This is the defect mechanism behind the theorem.

6.2 The all-negative baseline

If $Q_i = -1$ for every i, choose the lift $\tau_i = (-1)^i$. Let S be the bilateral shift and set

```
C=S+S^(-1),        E=S^2+S^(-2),        D=diag((-1)^i).
```

Then $A = C + DE$, while $CD = -DC$ and $ED = DE$. Hence the cross terms cancel:

`A^2=C^2+E^2=4I+S^2+S^(-2)+S^4+S^(-4).` (6.1)

On the Fourier mode $S = e^{i\theta}$, the symbol of (6.1) is

`4+2cos(2theta)+2cos(4theta)<=8.` (6.2)

Equality holds at $\theta = 0$. Therefore the all-negative phase has $R = 8$.

6.3 The first three moments at period eight

For an eight-periodic word, equations (2.9)-(2.10) give the signed closed-walk moments. The diagonal of A^2 is four at each of eight residues, so

`M_1=32.` (6.3)

Since A^2 is Hermitian,

`M_2=tr(A^4)=sum_(i,j)|(A^2)_(i,j)|^2.`

The diagonal and even-displacement terms contribute 160. Each positive flux opens four oriented odd-displacement entries, each of square four, so

`M_2=160+16d.` (6.4)

For $M_3 = \mathrm{tr}((A^2)^3)$, collect the local closed products according to whether they use no activated site, one activated site, an adjacent activated pair, or a distance-two activated pair. Direct multiplication of the displayed A^2 row gives respectively the coefficients 944, 168, 96, and 48:

`M_3=944+168d+96a+48b.` (6.5)

An independent derivation of (6.5), valid for every period, is given in Section 7 by enumerating the 430 closed step words of length six. Thus no spectral approximation enters (6.3)-(6.5).

The second excess is

`F_2=M_3-8M_2=-336+40d+96a+48b.` (6.6)

6.4 Four or more defects

Because $\prod_i Q_i = 1$, the defect count d is even.

Suppose $d = 4$. Write the four positive cyclic gaps as positive integers summing to eight. If at least two gaps are one, then $2a \geq 4$. If none is one, all four gaps equal two and $b = 4$. If exactly one is one, the other gaps are $2, 2, 3$, so again $2a + b = 4$. Hence always $2a + b \geq 4$, and (6.6) gives

`F_2=-176+48(2a+b)>=16>0.`

If $d = 6$, the two negative positions destroy at most four of the eight adjacent positive-positive edges, so $a \geq 4$; then $F_2 \geq 288 > 0$. If $d = 8$, (6.3)-(6.4) give

`F_1=M_2-8M_1=32>0.`

By Lemma 2.3, every phase with at least four defects has $R(Q) > 8$.

6.5 Two defects

Let $d = 2$ and let $s \in 1, 2, 3, 4$ be the smaller cyclic separation of the two defects. Translation and reflection make s a complete orbit invariant. Exact closed-walk enumeration gives the following first positive excesses:

separation s	first positive excess	value
1	F_4	5,504
2	F_6	64,336
3	F_9	2,872,096

We indicate how these are certified. Starting at each of the eight residues, enumerate all words of length $2k$ over steps $-2, -1, +1, +2$ whose total displacement is zero, multiply the corresponding signs from (2.1), and sum. This produces the integer M_k; subtraction gives F_k. For $s = 1, 2, 3$, the first positive values are exactly those in the table, so Lemma 2.3 gives $R(Q) > 8$.

For $s = 4$, the word is a translate of

```
Q_*=(+,-,-,-,+,-,-,-).
```

Section 5 proves $R(Q_*) = \eta < 8$. No conclusion is drawn from the negative values of its first nine excesses.

6.6 Proof of the trichotomy

The legal defect counts are $0, 2, 4, 6, 8$. Section 6.2 proves $R = 8$ when $d = 0$. Section 6.4 proves $R > 8$ for $d \geq 4$. Section 6.5 proves $R > 8$ for two-defect separations one, two, and three, while separation four is exactly the target orbit and has $R = \eta < 8$. These cases are disjoint and exhaustive, which proves Theorem D. □

The orbit of Q_* contains four flux words, corresponding to the four antipodal pairs. Thus the minimizer is unique modulo translation and reflection. The all-negative word is the only equality case at eight and is therefore the unique runner-up orbit.

6.7 Chiral structure of the target

The target lift may be written

```
tau=(+,-,+,-,-,+,-,+),        tau_(i+4)=-tau_i.                        (6.7)
```

Let $T_4(z)$ be translation by four sites on the Bloch fiber and let $D = \mathrm{diag}((-1)^i)$. On a general fiber,

```
(D T_4(z))^2=zI.
```

Choose xi with $xi^2 = z$ and define $J_z = xi^{-1}DT_4(z)$. Substitution of (6.7) in the four transitions gives

```
J_z^2=I,        J_z H(z)J_z^(-1)=-H(z).                                (6.8)
```

Both eigenspaces of J_z have dimension four. In a chiral basis,

```
H(z)=[0 B; C 0],        H(z)^2=[BC 0; 0 CB].                           (6.9)
```

On the unit circle $C = B*$. Equation (6.9) explains the even characteristic polynomial and the four-dimensional squared-band reduction. It does not by itself prove optimality: the legal chiral words form two dihedral orbits, one with two defects and one with six, and the latter lies above eight by Section 6.4.

7 General-Period Closed-Walk Obstructions

The local square identity in Section 6 does not depend on the period. We now derive the first three moments for every legal periodic phase and prove Theorem E.

7.1 From closed walks to flux monomials

The possible steps of (2.1) are $-2, -1, +1, +2$. A step of length one has weight one. A $+2$ step from j has weight τ_j, and a -2 step from j has weight τ_{j-2}.

Consider a closed step word and cancel repeated τ factors modulo two. The remaining indices occur in even number; write them as

```
r_1<r_2<...<r_(2s).
```

Using $Q_h = \tau_h \tau_{h+1}$ and telescoping,

```
product_j tau_(r_j)
 =product_(ell=1)^s product_(h=r_(2ell-1))^(r_(2ell)-1) Q_h.    (7.1)
```

Thus every signed closed walk becomes a monomial in interval products of Q. The identity is on the integer lattice, before reduction modulo a period, so it also handles short-cell collisions once cyclic indices are imposed.

7.2 Exact closed-word collection

For completeness, the collection has the following finite dynamic-programming recurrence on the integer lattice. For a starting residue r, set

```
W_0^(r)(j)=1 if j=r, and 0 otherwise,
W_(ell+1)^(r)(j)=sum_(delta in {-2,-1,1,2})
                 W_ell^(r)(j-delta) w(j-delta,delta),           (7.2)
```

where

```
w(i,-1)=w(i,+1)=1,   w(i,+2)=tau_i,   w(i,-2)=tau_(i-2).
```

After every multiplication, pairs of equal τ factors are cancelled and the remaining factors are converted to Q interval products by (7.1). Then

```
M_k=sum_(r=0)^(p-1) W_(2k)^(r)(r).                              (7.3)
```

Thus every coefficient below is obtained by additions and multiplications in the integral group algebra of the free sign variables; no floating-point spectral calculation enters the collection.

There are respectively 4, 36, and 430 closed step words of lengths two, four, and six. Collecting their flux monomials up to translation gives

length	contribution from one starting residue
2	4
4	$28 + 8Q_i$
6	$238 + 156Q_i + 24Q_iQ_{i+1} + 12Q_iQ_{i+2}$

We briefly justify the collection. At length two, each of the four steps is followed by its reverse. At length four, the 36 zero-sum words reduce via (7.1) to 28 constant terms and eight translates of a single Q_i. At length six, applying (7.1) to the 430 zero-sum words and grouping equal translated interval products leaves only the four displayed monomial types. This is a finite symbolic identity in the free Q variables, not a period-by-period numerical observation.

Summing over the p starting residues yields

```
M_1=4p,
M_2=28p+8 sum_i Q_i,
M_3=238p+156 sum_i Q_i
        +24 sum_i Q_iQ_(i+1)+12 sum_i Q_iQ_(i+2).               (7.4)
```

Let $I_i = (1 + Q_i)/2$. Then

```
d=sum_i I_i,        a=sum_i I_iI_(i+1),        b=sum_i I_iI_(i+2).
```

Substitute $Q_i = 2I_i - 1$ in (7.4) and collect. The result is

```
M_1=4p,
M_2=20p+16d,
M_3=118p+168d+96a+48b.                                            (7.5)
```

Because all sums are cyclic, (7.5) remains valid for $p = 1, 2, 3, 4$, even when the offsets one and two coincide modulo p; multiplicities are retained by the Laurent-matrix construction.

7.3 Necessary conditions at the eight-barrier

From (7.5),

```
F_1=M_2-8M_1=16d-12p,
F_2=M_3-8M_2=-42p+40d+96a+48b.                                    (7.6)
```

If $R(Q) \leq 8$, Lemma 2.3 requires $F_1 \leq 0$ and $F_2 \leq 0$. Therefore

```
d<=3p/4,
40d+96a+48b<=42p.                                                  (7.7)
```

This proves Theorem E. □

The 4, 36, and 430 word collections are a finite computer-assisted symbolic calculation. Recurrences (7.2)-(7.3) specify the complete proof algorithm, and the exact checker named in Appendix C reproduces the grouped table, formulas (7.4)-(7.6), and the period-eight values $F_4 = 5504$, $F_6 = 64336$, and $F_9 = 2872096$ used in Theorem D.

The first inequality limits defect density. The second also penalizes local clustering, assigning greater weight to adjacent pairs than to distance-two pairs. The converse is false in general: satisfying (7.7), or observing finitely many nonpositive F_k, does not establish $R(Q) \leq 8$.

7.4 Relation to the period-eight mechanism

At $p = 8$, equations (7.5)-(7.6) specialize to (6.3)-(6.6). For high defect density the first two excesses already cross the barrier. Low-density phases can evade these short tests, and the two-defect separation hierarchy in Section 6 shows that progressively longer closed walks are then required. This motivates the adaptive hierarchy used in the bounded low-period theorem.

8 The Low-Period Spectral Frontier

We prove Theorem F by combining complete orbit enumeration with a compressed set of exact spectral exclusions. Its domain is the set of infinite operators (2.1) whose Hamilton-gauge triangle word τ has a least positive period $p \leq 16$; the minimized quantity is the squared infinite-volume radius (2.5), not a finite-holonomy spectrum. A word displayed in a repeated cell is kept only for complete certificate accounting and is identified with its primitive phase by zone folding. The theorem does not extrapolate beyond this domain.

8.1 Complete phase space

For each displayed cell length $1 \leq p \leq 16$, consider the legal flux words

```
Q in {+1,-1}^p,        product_i Q_i=1,                            (8.1)
```

modulo rotations and reflections. Explicit orbit partition and the independent Burnside calculation of Appendix A give the counts

```
1,2,2,4,4,8,9,18,23,44,63,122,190,362,612,1162,                    (8.2)
```

whose sum is 2,626. Every periodic phase of primitive τ period at most 16 appears in the list when written in its primitive cell. Translation, reflection, global edge-sign negation, and cell repetition have the spectral equivalences proved in Lemmas 2.1-2.2.

8.2 Exact exclusion scheme

For every representative compute the moments (2.9). Lemma 2.3 supplies the valid implication

```
F_k>0 ==> R(Q)>8>eta.                                                   (8.3)
```

An adaptive exact closed-walk calculation through $F_6$4 finds a first positive excess for 2,611 representatives. The first-positive indices range from 1 to 64; in particular, two near-barrier classes are first detected only at $F_4$8 and $F_6$4. This calculation uses integer closed-walk sums, so the sign of every excess is exact.

Fifteen representatives remain. Eight are displayed even-period repetitions of the all-negative phase. Equation (6.1) proves $R = 8 > \eta$ for all of them. Two are

```
p=8:  Q=00010001,
p=16: Q=0001000100010001,                                          (8.4)
```

where zero denotes negative flux and one positive flux. Lemma 2.2 shows that the second row is the doubled-cell representation of the first, not a second phase; Section 5 gives their common value $R = \eta$.

The remaining five representatives are

p	canonical Q
10	0000010001
12	000000010001
14	00000000010001
14	00010010001001
16	0000000000010001

For each row choose $z \in +1, -1$ and a stored nonzero integer vector v. Let $H = H_Q(z)$ and

```
r=(v^T H^2 v)/(v^T v).                                             (8.5)
```

All arithmetic in (8.5) is integral before the final rational division. The five certificates first verify $r > 4$. Put

```
u=((r-4)^2-10)/2.
```

They then verify $u > 0$ and $u^2 > 5$. Since $r - 4 > 0$, these inequalities select the positive square-root branch and give

```
(r-4)^2>10+2sqrt(5),
r>4+sqrt(10+2sqrt(5))=eta.                                         (8.6)
```

The condition $r > 4$ is indispensable: the two inequalities involving u alone can also hold on the lower branch. Rayleigh's principle and (8.5)-(8.6) give $R(Q) \geq r > \eta$ for each residual representative.

8.3 Completeness accounting

The exact partition is

```
2611  moment-detected representatives,
   8  repeated all-negative representatives,
   5  endpoint-certified residual competitors,
   2  displayed target representations,
----
2626  total representatives.                                    (8.7)
```

The sets in (8.7) are disjoint, and Appendix A proves that the 2,626 representatives are the full legal orbit space. Every non-target phase in the domain therefore has $R(Q) > \eta$, while the two target rows are one primitive period-eight phase with $R(Q) = \eta$. This proves Theorem F. $\square$

The nearest observed competitor occurs at period ten, but numerical ordering within a non-target period is not part of the theorem. What is proved exactly is the strict comparison of every competitor with η.

9 Computer-Assisted Verification

Several principal results combine human arguments with finite exact computation. This section states precisely what is proved by computation, what is proved analytically, and how the two layers interact.

9.1 Proof classification

The Floquet direct sum, determinant identity, sharp positivity expansion, operator equivalences, cancellation baseline, moment barrier, conversion from the grouped moment identities to the defect inequalities, and chiral reduction are algebraic proofs presented in Sections 2 and 4-7.

The following assertions are finite computer-assisted theorems:

1. every switching class at each even order $8 \leq n \leq 30$ satisfies the conjectured bound;
2. the 2,626 legal flux/dihedral orbit representatives through displayed period 16 form a complete finite phase space;
3. the exact certificate partition used in Theorem F covers every one of those representatives;
4. the coefficient collection underlying (7.4)-(7.6), together with the stated F_4, F_6, and F_9 values, consists of exact computer-assisted symbolic identities reproduced by the general-period moment checker;
5. the closed-walk first-positive indices and five residual Rayleigh quotients are the exact outputs of integer or rational arithmetic.

Independent re-execution and independently written checkers strengthen confidence but are not additional hypotheses of the theorems.

9.2 Exactness of finite decisions

Floating-point eigenvalues are used only to locate likely maximizing fibers or to propose small rational vectors. A finite decision enters the proof only through one of the following exact forms:

- a rational Rayleigh inequality computed from an integer matrix and integer

vector;

- a positive-definiteness certificate from fraction-free or rational

elimination;

- a Sturm root-isolation certificate for an integer polynomial;
- an exact integer closed-walk excess;
- an exact orbit count and orbit-size sum.

Thus no strict theorem inequality depends on a floating tolerance.

9.3 Minimality computation

At $n = 8, ..., 20$, direct enumeration covers all 2^{n+1} switching classes. At $n = 22, ..., 30$, the computation enumerates canonical quadrilateral-flux bracelets, retains both holonomies, and attaches the exact dihedral orbit multiplicity. The represented switching-class total is checked against 2^{n+1}. The largest order therefore represents 2,147,483,648 switching classes by 17,929,600 canonical spectral states; it does not run 2.147 billion independent test cases.

Fresh deterministic regeneration was performed for the production orders:

n	canonical spectral states	chunks
24	353,812	28
26	1,299,064	76
28	4,810,472	250
30	17,929,600	908

The regenerated ordered input digests, ordered certificate digests, shell counts, represented totals, and terminal checkpoint chains agree with the original computations; the mathematical mismatch count is zero. Integrity replay of committed checkpoints is distinguished from this regeneration: it authenticates stored execution records but does not recreate every per-state inequality.

9.4 Low-period computation

For each $1 \leq p \leq 16$, one route explicitly partitions all legal flux words into dihedral orbits. A second route uses Burnside's lemma and permutation-cycle parities. The representative set, not merely its cardinality, is compared with an independently generated canonical set; deterministic orbit identifiers are bound to lexicographic order. Each stored row is then checked for legal parity, canonicality, orbit size, τ lift, primitive periods, operator equivalences, and an exact certificate.

The resulting 2,626-row table has no missing or duplicate canonical representative. The exact exclusion partition (8.7) has no overlap and no unresolved row.

9.5 Reproducible environment and trust limits

The submitted proof artifact is pinned to the public immutable snapshot

`https://github.com/whzy3185/math/tree/`
`c81be34a3b12a7ac47adbb4499c475df7bf4fc04`

The machine-readable manifest `research/reproducibility/target_a_submission_artifact_manifest.json` binds Theorems A and F to the SHA-256 hashes of the checker source, pinned requirements, order-24 through order-30 checkpoint chains, regeneration summary, and complete 2,626-row low-period table. For Theorem F the table contains every canonical representative and its exact decision. For Theorem A, the committed production checkpoints provide per-chunk integrity replay at orders 24 through 30, while the deterministic search program provides the full per-state mathematical regeneration route at every order. The compact artifact through order 22 is an aggregate attestation, not a substitute for that regeneration.

The reference environment is Python 3.12.13 with NumPy 2.3.5, SymPy 1.14.0, and pytest 9.1.1. A machine-readable requirements file pins the Python packages. The default suite and the three explicitly enabled slow generator audits can therefore be run from a clean Python 3.12 environment using repository-relative commands listed in Appendix C.

Two residual trust boundaries remain and are disclosed rather than hidden. For orders through 22, the compact canonical minimality artifact authenticates aggregate historical records; a fresh exact rerun has also been performed, but the repository does not archive an independently replayable vector for every historical state. For orders 26, 28, and 30, independent generator audits check Burnside totals, shell totals, represented-size sums, order, and parity, while recordwise equality against a separately implemented generator has been carried through order 24. These are execution-trust limits of the exhaustive part, not unreported numerical tolerances.

10 Discussion and Open Problems

The counterexample is governed by a simple local event that becomes visible only after squaring the operator. A negative quadrilateral flux removes its odd-displacement couplings from A^2, whereas a positive flux activates them with amplitude two. The all-negative phase is therefore a natural cancellation baseline at squared radius eight. A pair of positive defects can remain below that barrier only when it is antipodal in an eight-site cell; all other period-eight geometries create enough signed closed-walk mass to cross the barrier.

This mechanism explains why the conjecture survives many small finite orders. The target phase has an infinite-volume edge slightly below eight, while the twisted conjectured threshold increases toward eight. Their crossing is first realized at order 32. The same period-eight word then works at every larger multiple of eight because its Floquet spectrum is controlled uniformly over the full unit circle.

The arbitrary-period moment identities show that the defect picture is not an artifact of period eight. Any phase at or below the eight-barrier must have defect density at most $3/4$, and its adjacent and distance-two defect clusters must satisfy a stronger weighted inequality. These constraints are only necessary. The low-period frontier illustrates why: sparse near-barrier geometries can require closed walks of length up to 130 before a positive excess appears, and five bounded-period competitors are more efficiently excluded by endpoint Rayleigh vectors.

The computer-assisted results have two different functions. The finite search through order 30 establishes minimality of the counterexample. The bounded periodic classification establishes a structural frontier around the target. Neither computation proves an unbounded optimization theorem.

The current public-status check found no direct public resolution in the source paper, the author's listed follow-up materials, or the narrow direct searches performed on 20 August 2026. That assessment should be refreshed before submission and should not be read as an absolute priority statement.

The work leaves several natural questions.

Problem 1. Is the target phase globally optimal among periodic phases of arbitrary primitive period?

Problem 2. Is there a meaningful infinite-volume class containing nonperiodic signings in which the target remains optimal?

Problem 3. Which even orders not divisible by eight admit counterexamples, and what is the exact finite minimum at those orders?

Problem 4. Can the hierarchy $F_k = M_{k+1} - 8M_k$ be converted into a finite structural classification at arbitrary period, or do genuinely new local motifs appear at every scale?

Problem 5. Is there a flux-phase variational principle that explains the antipodal defect geometry without finite orbit classification?

Answers to these questions would move the present bounded and necessary results toward an unbounded flux-phase theorem. They are not assumed anywhere in this paper.

A Quotient and Orbit Completeness

This appendix supplies the finite coverage arguments used in Theorems A and F.

A.1 Switching coordinates

For $G_n = C_n(1, 2)$, the n triangle cycles together with the step-one Hamilton cycle form a basis of the $(n + 1)$-dimensional cycle space. Hence each switching class is represented uniquely by

```
(tau_0,...,tau_(n-1),alpha) in {+1,-1}^(n+1).                    (A.1)
```

For even n, global edge-sign negation negates every triangle flux and preserves α; it also replaces A by $-A$, so spectral radius is unchanged. The adjacent products $Q_i = \tau_i\tau_{i+1}$ are unchanged by this edge-sign negation. Legal Q words have product $+1$, and each has exactly the two lifts τ and $-\tau$. Therefore a pair (Q, α) represents precisely this spectrally equivalent pair of switching classes.

The dihedral group acts by graph automorphisms. If a canonical Q orbit has size s, retaining both values of α represents $4s$ switching classes: s geometric words, two τ lifts, and two holonomies. Summing over all legal Q orbits gives

```
sum_orbits 4s = 4*2^(n-1)=2^(n+1),                                (A.2)
```

which is exactly the switching-class count. Thus the quotient loses no class.

A.2 Burnside count for legal flux words

Let a permutation g of the p cyclic positions have cycle lengths $ell_1, ..., ell_c$. A word fixed by g is constant on each cycle, so it is specified by signs $epsilon_1, ..., epsilon_c$. Its total product is

```
product_(j=1)^c epsilon_j^(ell_j).                                 (A.3)
```

If at least one ell_j is odd, exactly half of the 2^c assignments make (A.3) positive, so g fixes 2^{c-1} legal words. If every ell_j is even, then (A.3) is always positive and g fixes all 2^c assignments. Burnside's lemma therefore gives

```
N_p=(1/(2p)) sum_(g in D_p)
    {2^(c(g)-1), if g has an odd cycle;
     2^c(g),     if all cycles of g are even}.                (A.4)
```

Evaluating the cycle decompositions of the p rotations and p reflections gives:

p	legal words	dihedral orbits
1	1	1
2	2	2
3	4	2
4	8	4
5	16	4
6	32	8
7	64	9
8	128	18
9	256	23
10	512	44
11	1,024	63
12	2,048	122
13	4,096	190
14	8,192	362
15	16,384	612
16	32,768	1,162

Their orbit total is 2,626.

A.3 Explicit representative-set equality

The bounded frontier uses more than the counts in (A.4). For each p, form all 2^{p-1} legal words by choosing the first $p-1$ entries and forcing the last entry to make the product positive. Replace each word by the lexicographically least of its rotations and reflected rotations. The set of resulting canonical words is compared exactly with the stored table's (p, Q) set, and the table identifiers are required to follow that lexicographic order. Equality of the sets excludes both an omitted orbit and a duplicate orbit hidden under a new identifier.

For every stored representative, a second check recomputes its dihedral image set, orbit size, periodic lift, primitive Q period, and primitive τ period. This proves that the two target rows in periods 8 and 16 are related by cell repetition and that no genuinely different primitive phase is removed.

A.4 Finite minimality coverage

At orders through 20, direct iteration over (A.1) covers every switching class. At the larger orders, canonical (Q, α) records are generated in defect shells. Each record stores its dihedral orbit size and therefore its represented multiplicity from (A.2). For every shell, the sum of these multiplicities is compared with the binomial parity count of legal words; the sum over shells is compared with 2^{n+1}. A terminal cursor proves that the canonical generator has exhausted its ordered domain.

At $n = 24$, a separately implemented visited-set generator and a fixed-weight necklace generator agree record for record on all 176,906 canonical flux records. At $n = 26, 28, 30$, independent Burnside and shell totals, represented size sums, ordering, and parity checks agree with production. This establishes the stated finite coverage subject to the execution-trust boundary disclosed in Section 9.

B Exact Classification and Residual Certificates

B.1 Period-eight orbit table

Use 0 for negative flux and 1 for positive flux. The 128 legal period-eight words form 18 dihedral orbits. The structural proof in Section 6 gives the following exact classification.

canonical Q	d	orbit size	primitive τ period	exact conclusion
00000000	0	1	2	$R = 8$
00000011	2	8	8	$R > 8$
00000101	2	8	8	$R > 8$
00001001	2	8	8	$R > 8$
00001111	4	8	8	$R > 8$
00010001	2	4	8	$R = \eta$
00010111	4	16	8	$R > 8$
00011011	4	8	8	$R > 8$
00100111	4	8	8	$R > 8$
00101011	4	16	8	$R > 8$
00101101	4	8	8	$R > 8$
00110011	4	4	4	$R > 8$
00111111	6	8	8	$R > 8$
01010101	4	2	4	$R > 8$
01011111	6	8	8	$R > 8$
01101111	6	8	8	$R > 8$
01110111	6	4	8	$R > 8$
11111111	8	1	1	$R > 8$

The orbit sizes sum to 128. The unique sub-eight orbit is 00010001, and the unique equality orbit at eight is 00000000.

B.2 Five low-period residual certificates

For every row, reconstruct the canonical triangle lift by setting $\tau_0 = +1$ and recursing through $\tau_{i+1} = Q_i \tau_i$. The vector coordinates are in the ordered fiber basis $(v_0, ..., v_{p-1})$ of Section 2.3. This fixes the matrix denoted by $H_Q(z)$ below without any gauge ambiguity.

For each row below, $H = H_Q(z)$ and the listed integer vector is denoted by v. Direct matrix multiplication gives the stated rational quotient $r = ||Hv||^2/||v||^2$.

p, Q	z	vector v	r
10, 0000010001	-1	$(3, 2, -3, 3, 0, 5, -5, 5, -5, 2)$	1066/135
12, 000000010001	1	$(-5, -5, -5, -1, -4, -2, -3, 0, -2, -2, 0, -4)$	1022/129
14, 00000000010001	1	$(-1, 2, 0, 2, -1, 3, -2, 3, -1, 4, -4, 4, -3, 0)$	119/15
14, 00010010001001	1	$(-4, -5, -4, 0, -2, 0, 1, 0, -2, 0, -4, -5, -4, -6)$	1270/159
16, 0000000000010001	1	$(-5, -5, -5, -1, -4, -2, -4, -1, -3, -1, -2, 0, -2, -2, 0, -4)$	1204/151

All five quotients exceed four. For the transformation

```
u=((r-4)^2-10)/2,
```

the exact comparison data are:

r	u	$u^2 - 5$
1066/135	47213/18225	568314244/332150625
1022/129	44813/16641	623590564/276922881
119/15	1231/450	502861/202500
1270/159	74573/25281	2365487524/639128961
1204/151	65995/22801	1755912020/519885601

Every entry in the last two columns is positive. Because $r > 4$, the branch argument (8.6) proves $r > \eta$, and Rayleigh's principle proves $R(Q) > \eta$. These five computations are the only individual endpoint certificates needed by the compressed proof of Theorem F.

The other triangle lift is obtained from the canonical one by the conjugacy and fiber reparametrization of Lemma 2.1; transporting v through that unitary map gives the same quotient and conclusion.

B.3 Exact two-defect moment table

For a period-eight word with two positive fluxes at cyclic separation s, the dynamic program (7.2)-(7.3) gives the following exact moments and excesses. The table includes every value needed to identify the first positive excess; only the positive entries are used to infer $R > 8$.

s	sequence	$k = 1, ..., 10$
1	M_k	$32, 192, 1376, 10976, 93312, 823920, 7447136, 68348832, 633982976, 5926419872$
1	F_k, $k = 1, ..., 9$	$-64, -160, -32, 5504, 77424, 855776, 8771744, 87192320, 854556064$
2	M_k	$32, 192, 1328, 9888, 76832, 612624, 4965328, 40683296, 335872832, 2788389472$
2	F_k, $k = 1, ..., 9$	$-64, -208, -736, -2272, -2032, 64336, 960672, 10406464, 101406816$
3	M_k	$32, 192, 1280, 9056, 66592, 503088, 3877920, 30363808, 240761792, 1928966432$
3	F_k, $k = 1, ..., 9$	$-64, -256, -1184, -5856, -29648, -146784, -659552, -2148672, 2872096$
4	M_k	$32, 192, 1280, 8928, 63872, 464496, 3417152, 25354656, 189356672, 1421345952$
4	F_k, $k = 1, ..., 9$	$-64, -256, -1312, -7552, -46480, -298816, -1982560, -13480576, -93507424$

Thus separations one, two, and three cross the barrier first at F_4, F_6, and F_9, respectively. The nonpositive displayed values for separation four are not used in the converse direction; that case is settled by the exact Floquet theorem of Section 5.

B.4 The order-32 positive-definiteness cross-check

For the witness (3.1), form $M = 1561I - 200A_3 2^2$. The exact certificate consists of the 32 positive pivots of rational LDL^T elimination. Fraction-free Bareiss elimination independently produces the leading principal minors, and the cumulative products of the LDL^T pivots agree with those minors. This double route verifies both arithmetic consistency and the hypothesis of Sylvester's criterion. The conclusion is the strict inequality $\rho(A_3 2)^2 < 1561/200$; no approximation to an eigenvalue is part of the certificate.

This elimination is an independent arithmetic cross-check, not the logical proof used by Proposition 3.1. The proposition follows directly from the displayed fiber matrix (4.2), determinant (4.5), positive expansion (4.6), and finite Floquet decomposition (4.3).

C Computational Protocol

C.1 Reference environment

The public proof artifact is the immutable Git snapshot

```
https://github.com/whzy3185/math/tree/
c81be34a3b12a7ac47adbb4499c475df7bf4fc04
```

and its submission binding is `research/reproducibility/target_a_submission_artifact_manifest.json`. The manifest records the artifact roles and SHA-256 hashes for Theorems A and F. Its verifier reads the named files from the pinned Git object, rather than silently accepting mutable working-tree copies. The snapshot contains all checker source, the complete 2,626-row low-period certificate table, the order-24 through order-30 checkpoint data, compact minimality artifacts, regeneration summary, and pinned dependencies. A DOI-bearing archival release is desirable at submission; the full commit identifier already makes this draft's public artifact immutable and content-addressed.

The reference interpreter and packages are:

```
Python 3.12.13
NumPy 2.3.5
SymPy 1.14.0
pytest 9.1.1
```

From a clean checkout with Python 3.12 available, a portable environment is created by

```
python3.12 -m venv .venv-target-a
.venv-target-a/bin/python -m pip install --upgrade pip
.venv-target-a/bin/python -m pip install \
  -r research/reproducibility/requirements-target-a.txt
```

All commands below are repository-relative.

C.2 Fast verification

The complete default regression suite is run by

```
.venv-target-a/bin/python -m pytest -q research/scripts
```

The principal theorem checkers may also be run separately:

```
.venv-target-a/bin/python research/scripts/verify_target_a_minimality_certificate.py
.venv-target-a/bin/python research/scripts/verify_target_a_period8_infinite_family.py
.venv-target-a/bin/python research/scripts/verify_target_a_period8_sharp_constant.py
.venv-target-a/bin/python research/scripts/verify_target_a_period8_structural_mechanism.
    py
.venv-target-a/bin/python research/scripts/verify_target_a_general_period_moments.py
.venv-target-a/bin/python research/scripts/verify_target_a_low_period_spectral_frontier.
    py
.venv-target-a/bin/python research/scripts/verify_target_a_low_period_structural_frontier
    .py
.venv-target-a/bin/python research/scripts/verify_target_a_periodic_operator_equivalences
    .py
.venv-target-a/bin/python research/scripts/verify_target_a_submission_artifact_manifest.
    py
```

Each command fails closed on a mismatched dependency, count, exact certificate, scope flag, or source digest.

C.3 Slow generator audits

The three generator audits skipped by the default suite are enabled explicitly:

```
TARGET_A_RUN_SLOW_GENERATOR_TESTS=1 \
.venv-target-a/bin/python -m pytest -q \
  research/scripts/test_target_a_direct_bracelets.py::DirectBraceletTests::
    test_direct_generator_burnside_n26 \
  research/scripts/test_target_a_direct_bracelets.py::DirectBraceletTests::
    test_direct_generator_burnside_n28 \
  research/scripts/test_target_a_direct_bracelets.py::DirectBraceletTests::
    test_direct_generator_burnside_n30
```

These tests independently check shell totals, Burnside counts, represented class totals, ordering, and parity. They are audits, not fresh spectral regeneration.

C.4 Regeneration versus integrity replay

Fresh regeneration starts from the canonical generator, reconstructs every spectral state, recomputes its exact decision, and writes a new checkpoint chain. Integrity replay reads an existing chain and verifies its hashes, counts, cursor, and aggregate records. Both are useful, but only the former re-executes the mathematical decision for every state.

For the production orders, choose an empty external directory and run

```
export OUT=/absolute/path/to/target-a-regeneration
mkdir -p "$OUT/checkpoints" "$OUT/results"
for n in 24 26 28 30; do
  .venv-target-a/bin/python research/scripts/target_a_minimality_search.py \
    --n "$n" \
    --checkpoint-dir "$OUT/checkpoints/n$n" \
    --output "$OUT/results/target_a_search_n$n.json"
done
```

An interrupted order is resumed by adding $--resume$ to its command. Each result JSON records the completed state and shell totals, ordered input and certificate hashes, terminal checkpoint-chain hash, elapsed time, and peak resident memory. A successful reference run on an Apple M5 using Python 3.12.13, NumPy 2.3.5, and SymPy 1.14.0 produced:

n	states	chunks	time (s)	peak RSS (MiB)	checkpoint disk	terminal chain SHA-256
24	353,812	28	25.3	75.1	1.1 MiB	2dde869aea5da4f040e67a4fef3e93b5f35f5fb42d75f6d48820439f418c83c1
26	1,299,064	76	117.2	84.4	3.1 MiB	c515350b8bea840c04448086fbc98523615364c05ca837553c93efb933bc0c4e
28	4,810,472	250	414.9	86.7	12 MiB	7ba200b05590b2a9c0ea1121f25f89ddf1d294d0c043417e69f78f764f6e8ee1
30	17,929,600	908	1542.3	135.2	43 MiB	b7fd264eece645eead187424152ae810a9ff940e37ffc5649b5ddf65aa31d59d

Runtime is hardware-dependent; the observed sequential total was about 35 minutes, peak per-process memory stayed below 150 MiB, and the four checkpoint directories occupied about 60 MiB. The compact committed reproduction summary also records the ordered input and ordered certificate hashes, so a fresh run can be compared without access to an off-repository timing log. The committed chains themselves are replayed by

```
.venv-target-a/bin/python research/scripts/target_a_checkpoint_replay.py \
  --n 24 26 28 30 \
  --output /tmp/target_a_checkpoint_replay.json
```

The production checkpoints are committed. The fresh regeneration chunks and full runtime logs are retained outside the repository; their compact manifest records state counts, chunk counts, ordered input and certificate digests, and terminal chain digests. Agreement of these logical fingerprints, rather than byte equality of timing metadata, defines mismatch zero.

C.5 Negative tests

Regression tests alter one logical field at a time and require rejection. They cover, among other faults, missing orbit representatives, duplicated canonical words, nondeterministic orbit identifiers, dependency digest drift, false primitive periods, altered Rayleigh numerators, the lower radical branch, incorrect moment direction, numeric previews promoted to proof, and forbidden all-period scope. This adversarial layer is important because many positive checks share the same exact arithmetic primitives.

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